SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I R SARVESH / 192521143 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R SARVESH

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models. (5 Marks)

7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

8. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

9. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity. (5 Marks)

10. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the

calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection. (5 Marks)

11. With the help of a neat diagram, explain the architecture of an **Artificial Neural Network (ANN)**. Describe the roles of input, hidden, and output layers, and explain how weights and biases are used in computation. Illustrate how forward propagation works with a simple example, and discuss how overfitting can be controlled using techniques such as dropout and regularization. (5 Marks)

12. Design a machine learning system for **weather forecasting** to predict rainfall using features such as temperature, humidity, wind speed, and atmospheric pressure. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time predictions. (5 Marks)

13. In the context of **Recurrent Neural Networks (RNNs)**, explain the architecture and how they handle sequential data. Describe the concept of hidden states and how information is passed through time steps. Illustrate with an example such as text prediction or time-series forecasting, and discuss challenges like vanishing gradients and possible solutions. (5 Marks)

14. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently. (5 Marks)

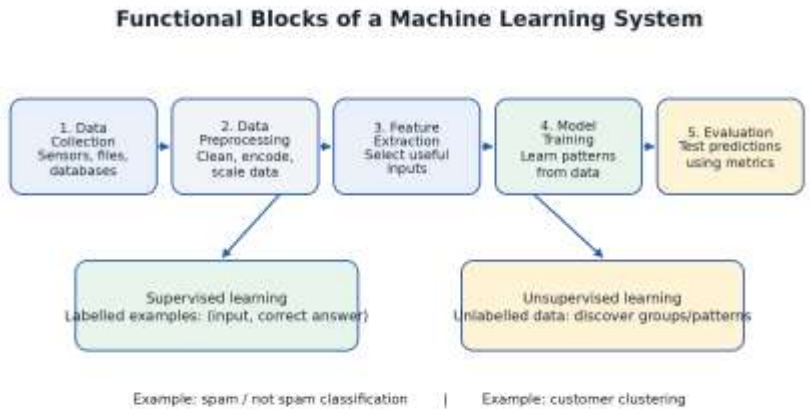
15. In the context of **classification models**, explain how performance evaluation is carried out using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Using a sample confusion matrix, demonstrate how each metric is calculated and discuss their importance in real-world applications such as spam detection and disease diagnosis. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. Functional Blocks of a Machine Learning System

A Machine Learning (ML) system learns patterns from data and uses them to make predictions or discover useful information.



Functional blocks:

1. Data Collection: Collects data from sensors, databases, files, or websites.
2. Data Preprocessing: Removes missing values, errors, and duplicate records and converts data into a suitable format.
3. Feature Extraction: Identifies important features required for learning.
4. Model Training: Trains an algorithm using the prepared data to learn patterns.
5. Evaluation: Tests the model using suitable metrics to measure its performance.

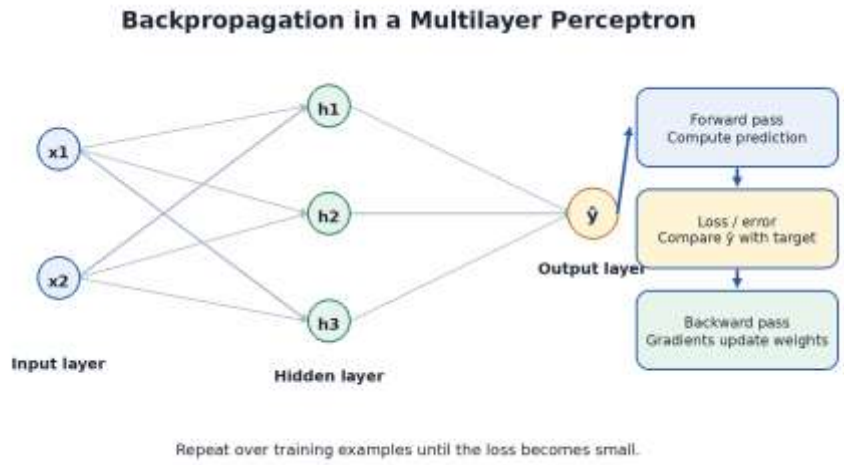
Types of learning:

- Supervised learning: Uses labelled data containing inputs and correct outputs. Example: classifying emails as spam or not spam.
- Unsupervised learning: Uses unlabelled data to identify hidden patterns or groups. Example: grouping customers based on purchasing behaviour.

Conclusion: These functional blocks help an ML system process data, learn patterns, and make accurate predictions.

Q3. Backpropagation Algorithm in a Multilayer Perceptron

Backpropagation is a learning algorithm used to reduce the error in a neural network by adjusting its weights and biases.



Steps involved:

1. Forward propagation: Inputs pass through the input and hidden layers to produce the predicted output.
2. Error calculation: The predicted output is compared with the target output.

Error = Target output - Predicted output

3. Backward propagation: The error is propagated backward through the network using the chain rule to calculate gradients.
4. Weight update: Weights are adjusted to reduce the error.

New weight = Old weight + (Learning rate*Error*Input)

5. Repeat: The process continues for multiple iterations until the error is sufficiently small.

Example:

If the target output is 1 and the predicted output is 0.6:

Error = 1 - 0.6 = 0.4

If learning rate = 0.1, input = 2, and old weight = 0.5:

Weight update = 0.1*0.4*2 = 0.08

New weight = 0.5 + 0.08 = 0.58

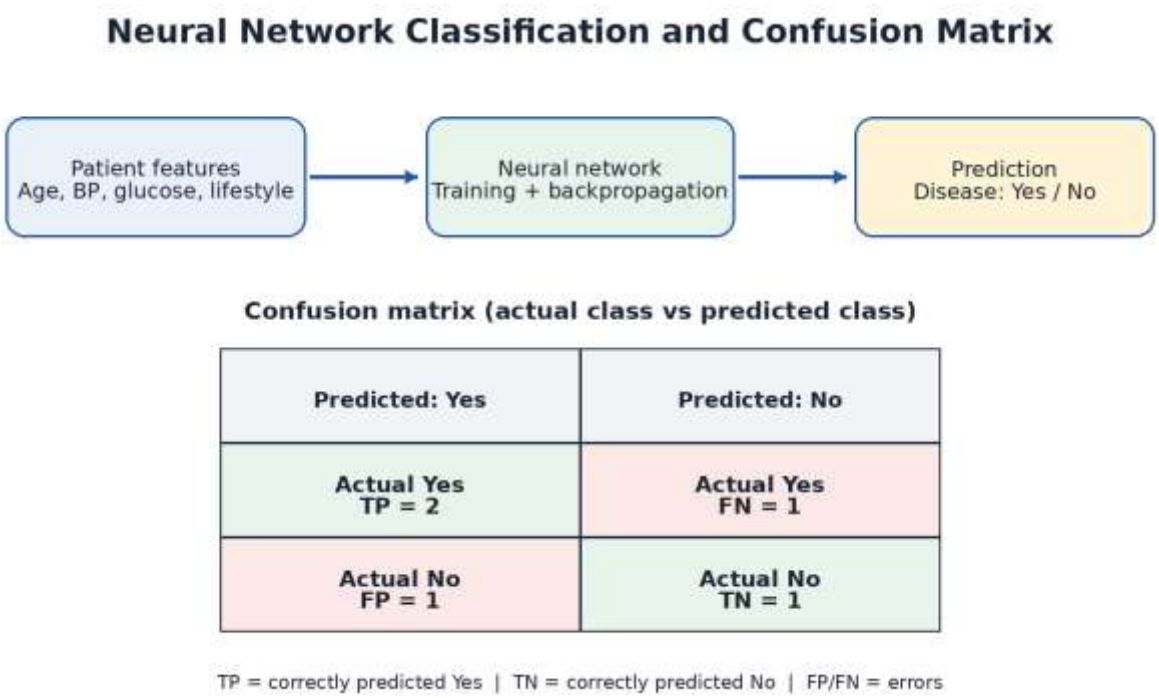
Role of learning rate and activation functions:

- Learning rate controls the size of each weight update.
- Activation functions such as sigmoid and ReLU help the network learn nonlinear patterns.

Conclusion: Backpropagation improves the network by updating weights and biases to minimize prediction error.

Q5. Neural Network Classification and Confusion Matrix

A neural network can classify patients as having a disease (Yes) or not having a disease (No) using features such as age, blood pressure, glucose level, and lifestyle risk.



Step 1: Training the model

1. Collect patient data and actual disease labels.
2. Preprocess the data and encode categorical values.
3. Select relevant features and divide the data into training and testing sets.
4. Train the neural network using forward propagation and backpropagation.
5. Compare predicted outputs with actual outputs to evaluate performance.

Step 2: Calculate the confusion matrix

From the given dataset:

Example	Actual	Predicted	Result
1	Yes	Yes	TP
2	Yes	No	FN
3	No	No	TN
4	Yes	Yes	TP
5	No	Yes	FP

Therefore:

- True Positive (TP) = 2
- True Negative (TN) = 1
- False Positive (FP) = 1
- False Negative (FN) = 1

Step 3: Evaluation metrics

Total = 2 + 1 + 1 + 1 = 5

Accuracy:

Accuracy = ((TP + TN)/Total)*100

Accuracy = ((2 + 1)/5)*100 = 60%

Precision:

Precision = (TP/(TP + FP))*100

Precision = (2/(2 + 1))*100 = 66.67%

Recall (Sensitivity):

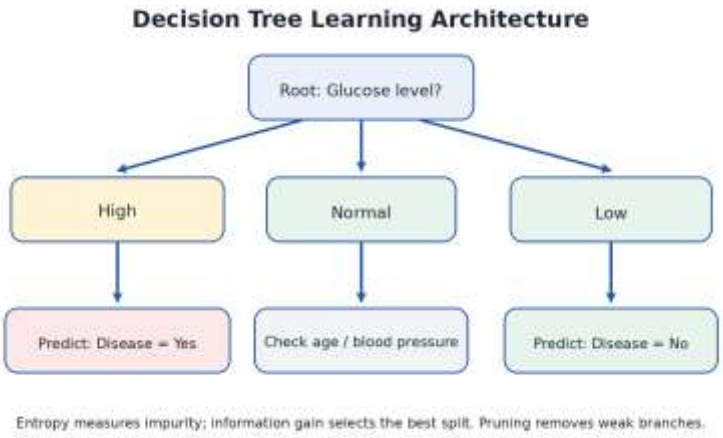
Recall = (TP/(TP + FN))*100

Recall = (2/(2 + 1))*100 = 66.67%

Conclusion: The model achieves 60% accuracy. Its performance needs improvement before it can be considered reliable for medical diagnosis. Preprocessing and selecting relevant features can improve prediction quality.

Q6. Decision Tree Learning System

A decision tree is a supervised machine learning algorithm that uses a tree-like structure to make predictions based on conditions applied to input features.



Main components:

1. Root node: The first feature used to split the dataset.
2. Decision nodes: Apply conditions to divide the data into smaller groups.
3. Branches: Represent the outcomes of conditions.
4. Leaf nodes: Represent the final predicted class or output.

Steps in tree construction:

1. Collect and preprocess the training dataset.
2. Calculate entropy to measure the impurity of the dataset.
3. Calculate information gain for the available features.
4. Select the feature with the highest information gain and split the dataset.
5. Repeat until a stopping condition is reached, producing the leaf nodes.

Formulas:

Entropy = -p1*log2(p1) - p2*log2(p2)

Information gain = Entropy(parent) - Weighted entropy(children)

Example:

A decision tree predicts whether a patient has a disease based on glucose level.

- If glucose is high, predict disease = Yes.
- If glucose is normal, check another feature such as blood pressure.
- If glucose is low, predict disease = No, according to the example tree.

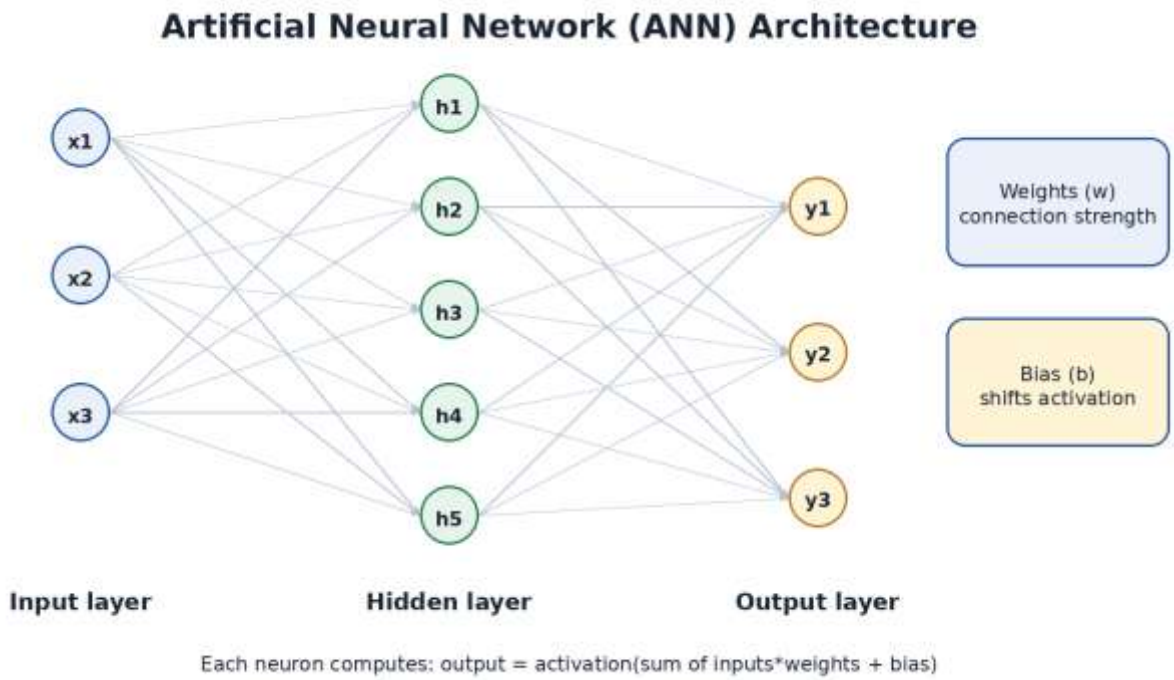
Pruning and overfitting:

Pruning removes unnecessary branches that fit the training data too closely. Overfitting can also be controlled by limiting tree depth, setting a minimum number of samples per leaf, and validating the model.

Conclusion: Decision trees are easy to understand and interpret. Entropy and information gain help select useful features, while pruning improves generalization.

Q11. Architecture of an Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is a machine learning model inspired by the human brain. It consists of interconnected artificial neurons that learn patterns from data.



Main layers:

1. Input layer: Receives input features such as study hours, attendance, or income.
2. Hidden layer: Processes inputs using weights, biases, and activation functions to learn patterns.
3. Output layer: Produces the final prediction, such as Pass/Fail or Yes/No.

Role of weights and biases:

- Weights: Determine the importance of each input.
- Bias: Helps adjust the neuron's activation threshold.
- Activation function: Converts the weighted sum into an output.

Forward propagation:

The input passes through the network to generate a prediction.

Net input = (x1*w1) + (x2*w2) + b

Output = Activation function(Net input)

Example:

Given x1 = 2, x2 = 1, w1 = 0.6, w2 = 0.4, and b = -0.5:

Net input = (2*0.6) + (1*0.4) - 0.5 = 1.1

Using a step activation function:

Output = 1, because 1.1 >= 0.

Therefore, the predicted class is Pass.

Controlling overfitting:

1. Dropout: Randomly disables some neurons during training to reduce dependency on particular neurons.
2. Regularization: Adds a penalty for large weights to discourage unnecessarily complex models.

Conclusion: ANNs learn complex relationships through interconnected layers. Forward propagation produces predictions, while backpropagation updates weights to improve accuracy.